Tim Keller, PhD

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For more details, particularly regarding projects, please refer to https://timkeller15.github.io.

SUMMARY

- Theoretical physicist with expertise in quantum optics, quantum computing, ultracold atomic gases, and high-performance computing for numerical simulations. Experienced in developing analytical models, optimal control strategies, and custom GPU-accelerated software for a broad variety of quantum systems from individual atoms to many-body cavity QED setups.
- Proven ability to independently design and execute complex research projects from theoretical modeling and large-scale simulation to data analysis and publication in peer-reviewed journals, while collaborating effectively across theoretical and experimental teams.

KEY AREAS OF EXPERTISE & TECHNICAL SKILLS

- Quantum Technologies | Applied Mathematics | Scientific Software Development | Technical Writing
- Julia | Python | CUDA | C++ | Slurm | Bash | Git | Matlab | Mathematica

EXPERIENCE

- **Junior Researcher (Assistant Professor)** **since Apr 2023**
  Waseda University, Tokyo, Japan
  - funded by JST Moonshot R&D Goal 6 Project: *Large-scale quantum hardware based on nanofiber cavity QED* led by Prof. Takao Aoki
  - developed theoretical models and numerical simulations to apply quantum computing performance benchmarks to neutral-atom qubits in a nanofiber cavity QED platform, guiding experimental optimization of quantum gate fidelities
  - designed a custom GPU-accelerated numerical FDTD solver for Maxwell's equations to model laser-induced trapping potentials near optical nanofibers and calculate fast and efficient atom-loading protocols
- **JSPS Postdoctoral Fellow** **Sep 2022 - Mar 2023**
  OIST, Okinawa, Japan
  - extended PhD work on many-body quantum dynamics and control of interacting quantum gases
  - supervised and mentored three PhD students and two research interns, providing training in numerical simulation and analytical techniques

FELLOWSHIPS & AWARDS

- **Research Fellowship for Young Scientists (DC2)** **Apr 2021 - Mar 2023**
  Japan Society for the Promotion of Science **Grant:** JSPS KAKENHI 21J10521
  Stipend: JPY 200 000 ($\sim$ USD 1800) per month & Research Grant of JPY 900 000 ($\sim$ USD 8200)
- **Physical Review Letters Editor's Suggestion** **Jan 2022**
  American Physical Society **Publication:** PRL **128**, 053401 (2022).
  Flagship journal with acceptance rate $< 25\%$. Only one in seven Letters highlighted as Editor's Suggestion due to their particular importance, innovation, and broad appeal.

EDUCATION

- **PhD in Physics** **Sep 2017 - Aug 2022**
  Thesis: Controlling Superfluid and Insulating States in Interacting Quantum Gases
  Okinawa Institute of Science and Technology (OIST), Japan **Advisor:** Prof. Thomas Busch
- **MSc in Physics** **Oct 2014 - Dec 2016**
  Saarland University, Germany **Advisor:** Prof. Giovanna Morigi
- **BSc in Physics** **Oct 2011 - Aug 2014**
  Saarland University, Germany **Advisor:** Prof. Giovanna Morigi

PROJECTS & PUBLICATIONS

(10) **Cavity and waveguide quantum electrodynamics with atoms and optical nanofibers**
Contribution: *K. Webb, S. Ruddell, T. Keller, J. Keloth, R. Gupta, M. Takahata, and T. Aoki*
Publication: Nano Convergence **13**, 35 (2026).

(9) **Hikari.jl - GPU-accelerated FDTD Maxwell equation solver for calculating laser-generated atomic trapping potentials close to optical nanofibers**
Contribution: *T. Keller* Code: Hikari.jl

(8) **Addressing requirements for crosstalk-free quantum-gate operation in many-body nanofiber cavity QED systems**
Contribution: *T. Keller, S. Kikura, R. Asaoka, Y. Suzuki, Y. Tokunaga, and T. Aoki*
Publication: J. Opt. Soc. Am. B **43**, 351-365 (2026). Code: AddressingRequirements.jl

(7) **Fermionization of a few-body Bose system immersed into a Bose-Einstein condensate**
Contribution: *T. Keller, T. Fogarty, and T. Busch*
Publication: SciPost Phys. **15**, 095 (2023). Code: Fermionization

(6) **Self-Pinning Transition of a Tonks-Girardeau Gas in a Bose-Einstein Condensate**
Contribution: *T. Keller, T. Fogarty, and T. Busch*
Publication: Phys. Rev. Lett. **128**, 053401 (2022). **(Editor's Suggestion)** Code: Self-Pinning

(5) **Adiabatic critical quantum metrology cannot reach the Heisenberg limit even when shortcuts to adiabaticity are applied**
Contribution: *K. Gietka, F. Metz, T. Keller, and J. Li*
Publication: Quantum **5**, 489 (2021).

(4) **Feshbach engine in the Thomas-Fermi regime**
Contribution: *T. Keller, T. Fogarty, J. Li, and T. Busch*
Publication: Phys. Rev. Research **2**, 033335 (2020). Code: Feshbach-Engine

(3) **Quenches across the self-organization transition in multimode cavities**
Contribution: *T. Keller, V. Torggler, S. Jäger, S. Schütz, H. Ritsch, and G. Morigi*
Publication: New J. Phys. **20** 025004 (2018). Code: MultimodeSelforganization.jl

(2) **Phases of cold atoms interacting via photon-mediated long-range forces**
Contribution: *T. Keller, S. Jäger, and G. Morigi*
Publication: J. Stat. Mech. 064002 (2017). Code: MultimodeSelforganization.jl

(1) **Probing the out-of-equilibrium dynamics of two interacting atoms**
Contribution: *T. Keller and T. Fogarty*
Publication: Phys. Rev. A **94**, 063620 (2016). Code: TwoAtomQuenches.jl

LANGUAGES

German: Native | **English:** Fluent (TOEFL Score 114/120) | **Japanese:** Intermediate (JLPT N2 Certificate)